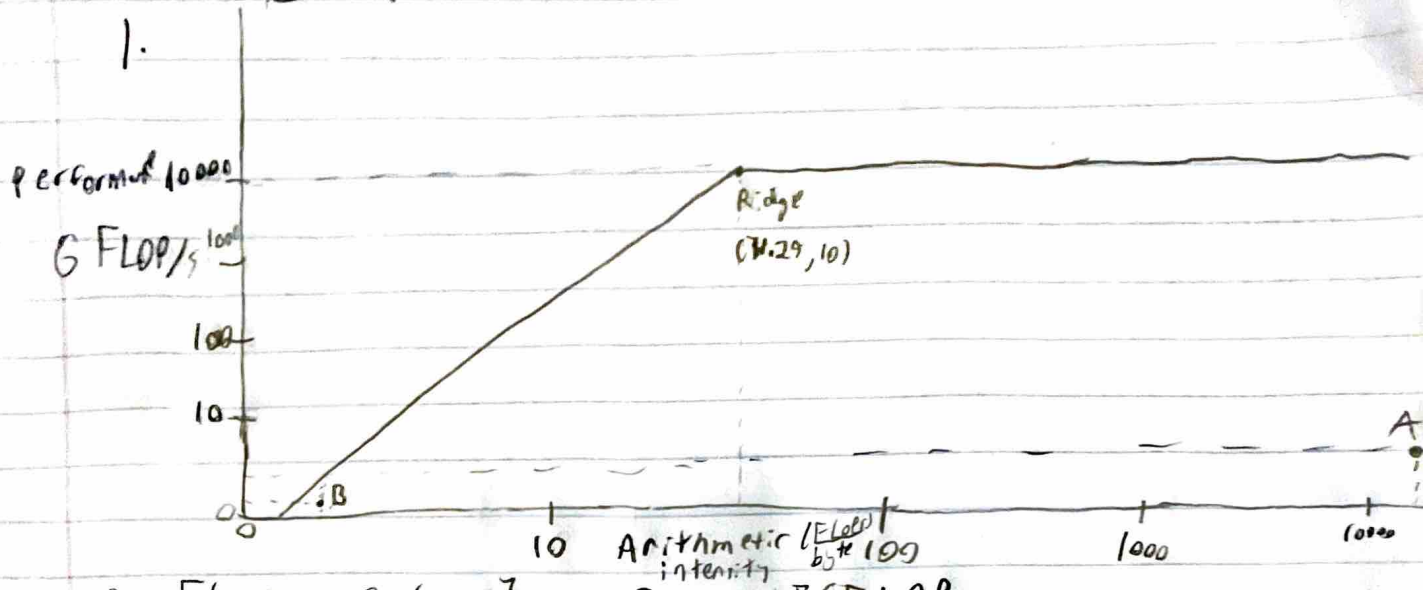
ECE 410 Codefest 2 CMAN

1.



2. FLOPs $= 2 \cdot 1024^3 \approx 2.14748$ GFLOP

bytes transferred $= 4 \cdot 3(1024) = 12288$ bytes

$$AI = \frac{2.14748 \text{ GFLOP}}{12288 \text{ byte}} = 174763$$

3. FLOPs $= 4.194304$ MFLOP

bytes transferred $= 4194304(2)(4)$
(assuming no cache reuse) $= 33554432$ bytes

$$AI = \frac{4.194304 \text{ MFLOP}}{33554432 \text{ byte}} = 0.125 \frac{\text{FLOPs}}{\text{byte}}$$

4. a) Kernel A: compute bound
Kernel B: memory bound

b) Kernel A: 10T FLOP
Kernel B: ~ 2.14 GFLOP

c) Improving memory transfer rate since Kernel A is heavily bottlenecked by it, more so than Kernel B's computational bottleneck.